

Do phytoestrogens reduce the risk of breast cancer and breast cancer recurrence? What clinicians need to know.

Oestrogen is an important determinant of breast cancer risk. Oestrogen-mimicking plant compounds called phytoestrogens can bind to oestrogen receptors and exert weak oestrogenic effects. Despite this activity, epidemiological studies suggest that the incidence of breast cancer is lower in countries where the intake of phytoestrogens is high, implying that these compounds may reduce breast cancer risk, and possibly have an impact on survival. Isoflavones and lignans are the most common phytoestrogens in the diet. In this article, we present findings from human observational and intervention studies related to both isoflavone and lignan exposure and breast cancer risk and survival. In addition, the clinical implications of these findings are examined in the light of a growing dietary supplement market. An increasing number of breast cancer patients seek to take supplements together with their standard treatment in the hope that these will either prevent recurrence or treat their menopausal symptoms. Observational studies suggest a protective effect of isoflavones on breast cancer risk and the case may be similar for increasing lignan consumption although evidence so far is inconsistent. In contrast, short-term intervention studies suggest a possible stimulatory effect on breast tissue raising concerns of possible adverse effects in breast cancer patients. However, owing to the dearth of human studies investigating effects on breast cancer recurrence and survival the role of phytoestrogens remains unclear. So far, not enough clear evidence exists on which to base guidelines for clinical use, although raising patient awareness of the uncertain effect of phytoestrogens is recommended.